

Review Article

A review of state-of-health estimation for lithium-ion battery packs

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ABSTRACT

With the rapid advancement of lithium-ion battery technology, the estimation of the state of health (SOH) of lithium-ion battery packs plays a crucial role in enhancing the safety and reliability of their operation. However, few researchers have reviewed SOH estimation for lithium-ion battery packs so far. In order to fill this gap, this paper will review and discuss the past proposed methods on SOH estimation for lithium-ion battery packs. This paper first introduces the working principle of lithium-ion battery packs and their degradation mechanisms at chemical and mechanical levels during continuous charging and discharging, and describes the impact of internal inconsistencies in battery packs on overall performance degradation. This is followed by a systematic summary of existing degradation state identification and SOH estimation methods and an analysis of the limitations of these methods in practical applications. Lastly, the difficulties currently encountered in estimating the SOH of Li-ion battery packs are discussed, along with potential directions for future research. It is hoped that this paper will be a useful reference for scholars working in relevant fields.

1. Introduction

The application of electric vehicles and energy storage power plants has become increasingly promising with the rapid growth of distributed energy storage and renewable energy, along with China's objective of "carbon peaking and carbon neutrality". According to the China Association of Automobile Manufacturers (CAAM), 9.495 million new energy cars were sold in China in 2023, while 9.587 million were produced. These numbers show annual increases of 35.8 % and 37.9 %, respectively. The market share increased to 31.6 %, and the market share reached 31.6 %, which was higher than the same period of the previous year, when the market share reached 5.9 % [1]. A crucial piece of technology for easing the energy problems electric cars confront is electrochemical energy storage [2]. The superior features of Li-ion batteries, such as their high energy density, low self-discharge rate, long operational lifespan, lack of memory effect, and environmental sustainability, have made them the dominant technology in power and energy storage systems when compared to other technologies like lead-acid and nickel-cadmium batteries. These advantages facilitate their extensive application in both civilian and military sectors [3–6].

With repeated cycles of charging and discharging, the irreversible electrochemical reactions within lithium-ion batteries increase internal resistance, gradually leading to a decline in capacity and performance. When the battery's performance deteriorates beyond a certain threshold,

issues such as electrolyte leakage and localized short circuits may occur, increasing the risk of safety incidents [7]. Zhang et al. [8] conducted statistics on the domestic new energy vehicle fire incidents between 2018 and 2020, from the data they collected, in 2018, there were a total of 22 new energy vehicle fire incidents, while in 2020 this data grew to 31, a total of 72 new energy vehicle fire incidents during the three-year period, and the new energy vehicle fire accidents showed an increasing trend, and these most of these accidents are caused by improper use of batteries. Therefore, the SOH estimation method for Li-ion battery packs has attracted extensive attention from scholars. Accordingly, there are many literatures reviewing the SOH estimation methods for lithium-ion batteries. For example, Literature [9] systematically summarizes and concludes the SOH prediction methods for lithium-ion batteries through machine learning. Literature [10] provides an in-depth exploration of three typical modeling approaches for SOH estimation and offers a comprehensive review of the various evaluation criteria employed. Additionally, it quantitatively assesses the performance of these methods from multiple perspectives. Literature [11] provides a systematic overview of SOH in lithium-ion batteries from two perspectives: characterization and estimation, and presents current applications for both short-term estimation and long-term prediction of SOH. At present, electric vehicles are a class of complex systems in the civil field that widely use lithium-ion batteries, for this point, literature [12] examines lithium-ion battery health state estimate and prediction techniques in

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the context of electric vehicle operation, elucidating the requirements and difficulties associated with health state estimation in these application situations.

Lithium-ion batteries are typically coupled in series or parallel combinations to produce battery packs in real-world applications, enabling them to meet system requirements for capacity, power output, voltage, and other performance criteria [13]. However, due to the inconsistency between single batteries in a lithium-ion battery pack, the performance of the battery pack is jointly affected by the performance of the internal batteries and the consistency between the batteries [14]. Typically, the performance degradation of a battery pack occurs more rapidly than that of individual cells, especially after the cells are assembled into a pack. Significant changes occur in its monitoring parameters, charging and discharging patterns, and other operational modes [15]. Consequently, in addition to estimating the SOH of individual Li-ion cells, it is imperative to thoroughly consider the composition and structure of the battery pack, along with cell inconsistencies and other characteristics. This will enable the development of more precise methods for estimating and predicting SOH in battery packs, thereby addressing the needs of real-world applications and guaranteeing the reliable functioning of the system. Especially in the context of electric vehicle applications, it is crucial to accurately assess the SOH of lithium-ion battery packs to ensure they function safely and reliably.

In summary, the operational process of Li-ion battery packs is more complex than that of individual Li-ion batteries. Following battery assembly in practical applications, the quantity and types of parameters that can be monitored in Li-ion battery packs differ from those in individual cells. Therefore, despite the availability of SOH estimation methods for individual batteries, significant research challenges remain in the assessment of SOH of Li-ion battery packs [16].

While extensive research has been conducted on individual cell SOH evaluation, the assessment methods for Li-ion battery packs remain relatively unexplored in current literature. This review addresses this research gap by comprehensively analyzing and synthesizing existing methodologies for Li-ion battery packs SOH estimation.

This paper provides valuable insights to battery researchers and specialists by exploring advanced techniques in SOH prediction and forecasting future trends for lithium-ion battery systems. Additionally, it aims to catalyze further advancements in new energy vehicle technology. The document is structured into six main sections. Initially, Section 2 delves into the degradation processes of lithium-ion batteries and their assemblies. Following this, Section 3 evaluates various methods for identifying degradation features within these battery packs, highlighting comparisons among these techniques. Section 4 elaborates on current SOH prediction strategies, incorporating statistical filters, data-centric models, and integrated methods. Section 5 tackles the prevalent obstacles in accurately predicting SOH for lithium-ion battery packs and offers targeted suggestions. A concise summary of the findings and perspectives is presented in the concluding section.

2. Degradation mechanism analysis of lithium-ion battery packs

The deterioration of Li-ion battery packs is a multifaceted process driven by a range of physical and chemical factors [17]. To clarify this process, the following paper will provide an in-depth introduction to the working principles of Li-ion batteries, along with a detailed examination of the degradation processes affecting individual Li-ion cells and entire battery packs.

2.1. How lithium-ion batteries work

Li-ion batteries function as rechargeable devices through the movement of lithium ions between positive and negative electrodes [18], as illustrated in Fig. 1. These energy storage systems consist of five essential components: positive electrode, negative electrode, separator, organic electrolyte, and current collector [19]. The separator maintains

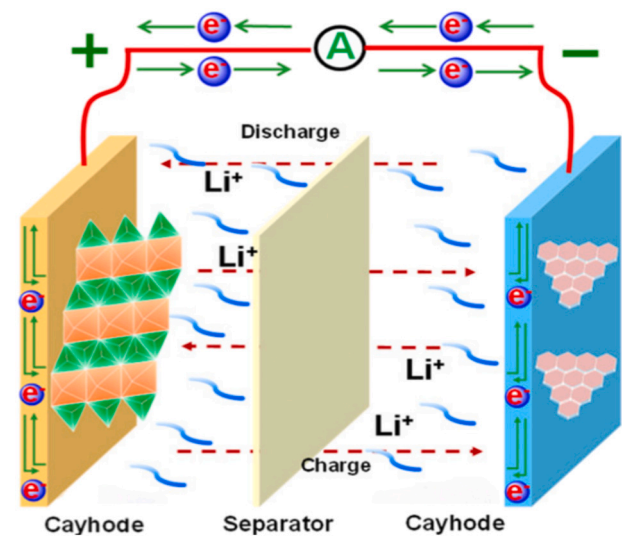
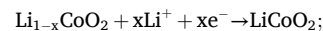


Fig. 1. Lithium-ion battery working diagram.

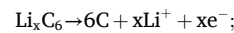
physical isolation between electrodes when immersed in electrolyte, while the battery shell provides protection for internal components. The positive electrode of lithium-ion batteries is usually composed of lithium-containing oxides, such as LiCoO_2 , LiMn_2O_4 , LiFePO_4 and so on [20]. The negative electrode, on the other hand, is mostly made of carbon materials [21]. During the charging process, lithium ions migrate from the positive electrode material through the separator and electrolyte, ultimately embedding themselves in the negative electrode material. This movement creates a lithium-enriched state at the negative electrode while depleting lithium content at the positive electrode. The discharge process reverses this migration: lithium ions release from the negative electrode, traverse the electrolyte and separator, and integrate into the positive electrode material, resulting in lithium enrichment at the positive electrode and depletion at the negative electrode. Take the common lithium cobalt cobaltate lithium-ion battery as an example (the positive pole, the negative pole is embedded lithium carbon material), In a chemical reaction, the positive and negative electrodes are charged and discharged according to the following formula:

(1) When the battery is charging:

Positive reaction:



Negative reaction:

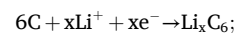


(2) When the battery is discharged:

Positive reaction:



Negative reaction:



Lithium ions flow back and forth between the positive and negative electrodes of lithium-ion batteries, which allows them to store and release electrical energy.

2.2. Degradation mechanism analysis of lithium-ion battery

As sophisticated electrochemical systems, Li-ion batteries experience performance changes due to multiple external factors: varied charging and discharging patterns, fluctuating ambient temperatures during operation, and non-uniform substance distribution during cyclic usage. These conditions trigger irreversible electrochemical reactions that manifest through diminished usable capacity, attenuated power output, reduced discharge efficiency, and elevated internal resistance [22]. The cumulative impact of these degradation mechanisms significantly influences the operational stability and reliability of various critical applications, including battery energy storage systems, electric vehicles, and aviation satellites [23].

The degradation mechanisms of lithium-ion cells can be categorized into two fundamental types based on their deterioration causes: mechanical and chemical degradation mechanisms [24]. Fig. 2 provides a comprehensive illustration of these degradation reasons in lithium-ion battery cells.

According to chemical principles, Li-ion batteries experience multiple degradation mechanisms that affect their performance [25]:

- (1) Electrolyte transformation: Throughout operational cycles, the electrolyte system undergoes complex decomposition and reduction reactions. These electrochemical processes progressively alter its fundamental composition, which subsequently leads to substantial modifications in the battery's electrochemical performance characteristics.
- (2) Solid electrolyte film formation and growth: A distinctive solid electrolyte interphase (SEI) layer gradually develops and

accumulates on the surfaces of both positive and negative electrodes. The continuous formation process and progressive growth of this interfacial film will influence the battery's long-term operational stability and functional durability.

- (3) Dissolution of active materials: During extended periods of electrochemical cycling, the functional materials embedded within both electrode structures may experience progressive disintegration. This gradual degradation process compromises the electrochemical system's reversible capacity and operational longevity.
- (4) Gas precipitation and loss of lithium ions: The precipitation of dissolved gasses in the electrolyte and the loss of lithium ions during the charging and discharging operations can both impair battery performance.

The mechanical degradation mechanism, distinct from chemical degradation pathways, centers on the volume changes of active material particles and consequent stress generation during lithium ion 'embedding' and 'de-embedding' processes at both electrodes. In real-world applications, these various degradation mechanisms interact across all battery components - cathode, anode, separator, electrolyte, and current collector - working in concert to diminish overall battery performance.

For lithium-ion batteries with various electrode types, such as lithium cobalt oxide, lithium iron phosphate, and lithium manganese oxide, the primary causes of degradation vary under different operating conditions—including charge and discharge rates, depth of discharge, and environmental temperature. And when batteries are connected in both series and parallel to form a battery pack, it results in different degradation mechanisms for each battery and increases inconsistency

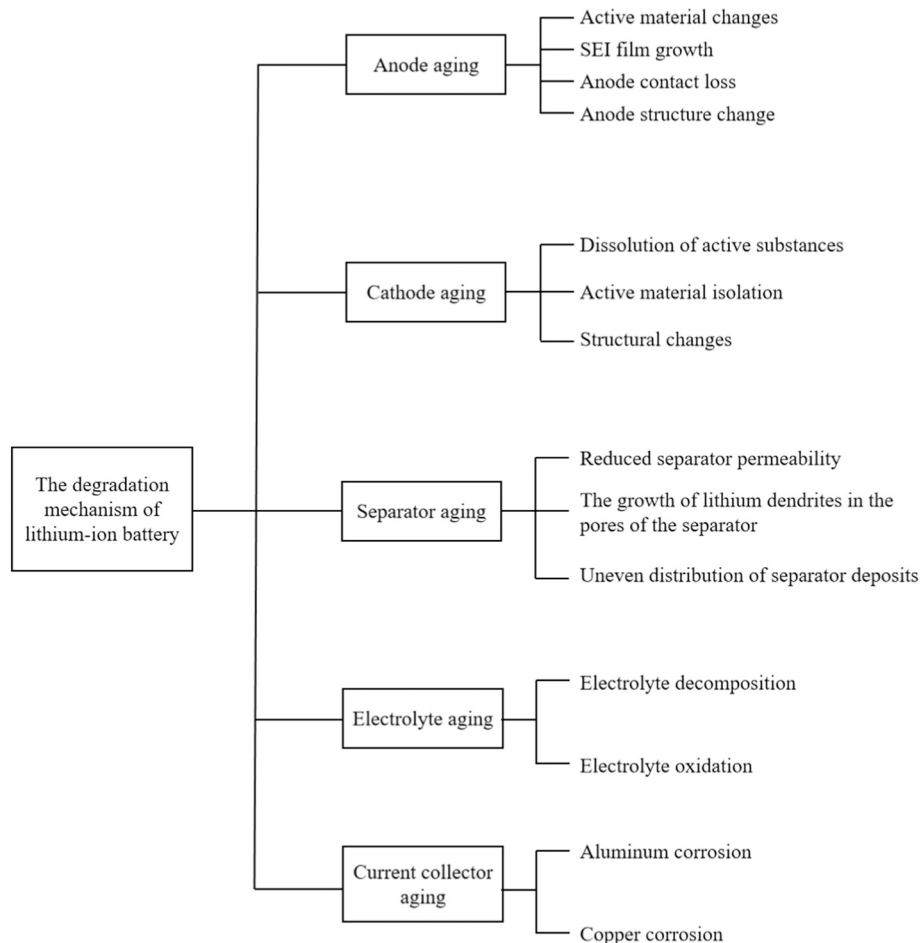


Fig. 2. Lithium-ion battery cell degradation mechanism.

among the batteries within the pack, ultimately impacting the pack's functionality and longevity.

2.3. The impact of fast charging on battery health

Fast charging technology, as a crucial means of enhancing battery convenience, has been widely implemented in new energy vehicles and portable electronic devices. Generally, electric vehicle charging processes exceeding 2 h are classified as standard slow charging, while those ranging from 0.5 to 2 h are categorized as fast charging [26]. However, high-rate charging processes significantly accelerate battery performance degradation, imposing more stringent requirements on battery health and safety.

Research indicates that during fast charging, high current density leads to lithium ion deposition on the graphite anode surface, potentially resulting in lithium dendrite formation. For instance, conventional fast charging protocols (such as CC/CV) can fully charge a battery within 6 min but induce substantial lithium plating, resulting in approximately 20 % capacity degradation after 50 cycles [27]. Simultaneously, repeated formation and breakdown of the SEI film consume active lithium ions, thereby reducing coulombic efficiency and capacity [26]. Furthermore, the high currents during fast charging cause internal battery temperature elevation, and excessive temperatures accelerate chemical reactions, consequently diminishing battery cycle life. Additionally, elevated electrolyte temperatures accelerate electrolyte decomposition while increasing the risks of flammability and corrosion phenomena. The synergistic effects between low temperature (e.g., -5°C) and fast charging are particularly pronounced: Kumar et al. [28] experimentally demonstrated that the lithium plating area ratio nearly doubles under low-temperature fast charging compared to room temperature conditions, with series resistance increasing by 73 % and charge transfer resistance rising by 16 %.

In conclusion, while fast charging technology offers convenience through accelerated charging processes, it exerts significant impacts on battery SOH. High charging rates lead to elevated battery temperatures, accelerating internal chemical reactions and consequently exacerbating capacity fade. Additionally, rapid charging may induce lithium ion deposition on the battery's negative electrode surface, forming lithium dendrites that cause anode degradation and impact battery lifespan.

2.4. Degradation mechanism analysis of lithium-ion battery packs

Li-ion battery packs are composed of multiple lithium-ion battery monomer by series, parallel or series-parallel combination to obtain higher voltage and enhanced capacity to fulfill the requirements of the equipment. Nevertheless, it is frequently challenging for battery packs to use the battery monomer's full potential in actual applications, which is highlighted here in the maximum charge/discharge capacity, charge/discharge efficiency, and other indicators [29]. This is mainly because in the manufacturing and production process, the production process, the production environment and other minor differences will lead to the initial state of each monomer between the existence of unavoidable

inconsistency [30]. At the same time, in the battery pack work process, due to the inconsistency of the capacity, internal resistance, charge state, etc. that may exist in the initial state of the battery monomer, the charging and discharging conditions between the monomer within the battery pack will also have a certain degree of variability [31]. These differences are specifically manifested in the uneven heat distribution within the battery pack [32], uneven current distribution between the monomer [33], uneven charging and discharging voltage intervals between the monomer, etc. [34], which will result in inconsistent deterioration of the performance of the individual cells within the battery pack, which will in turn exacerbate the inconsistency between the monomer of the Li-ion battery.

In series-connected lithium-ion battery packs, the 'barrel effect' [35] (depicted in Fig. 3) manifests throughout the charge-discharge cycle to protect cells from overcharging and overdischarge conditions. This phenomenon significantly influences the pack's operational parameters, including its utilizable capacity, energy conversion efficiency, and various performance metrics.

During both charging and discharging processes, parallel-connected lithium-ion batteries maintain equivalent terminal voltages. However, due to variations in the direct current internal resistance among individual cells, the current distribution through each battery differs [36].

In addition, there are inconsistencies in the heat production characteristics of battery monomer. After the battery cells are connected in series or parallel to form a battery pack, the temperature distribution within the pack may exhibit significant inconsistencies due to uneven charging and discharging intervals, variations in current distributions, and differing cooling conditions among the individual cells [37]. This inhomogeneity of temperature distribution will further exacerbate the performance degradation gap between the battery monomer, which will lead to further expansion of the inconsistency of the capacity, impedance, and other parameters of the battery monomer, and ultimately exacerbate the performance degradation of the battery pack.

In summary, a significant factor contributing to the decline in the performance of lithium-ion battery packs is the inconsistency among individual cells. Variations in initial parameters, operational states, conditions, and working environments among the cells can lead to discrepancies in their performance once the lithium-ion battery cells are assembled into a pack, as illustrated in Fig. 4. Performance degradation inconsistency expansion, thus further exacerbating the inconsistency between the monomer, forming a vicious cycle, accelerating the performance deterioration of Li-ion battery packs.

3. Health feature extraction of lithium-ion battery pack

The health feature extraction in lithium-ion batteries involves analyzing multiple parameters, including voltage, current, and temperature measurements from individual cells or complete battery packs. This process focuses on extracting parameters that demonstrate clear trend patterns, enabling comprehensive characterization of performance deterioration in Li-ion battery packs. Through systematic monitoring and analysis of these parameters, the degradation can be

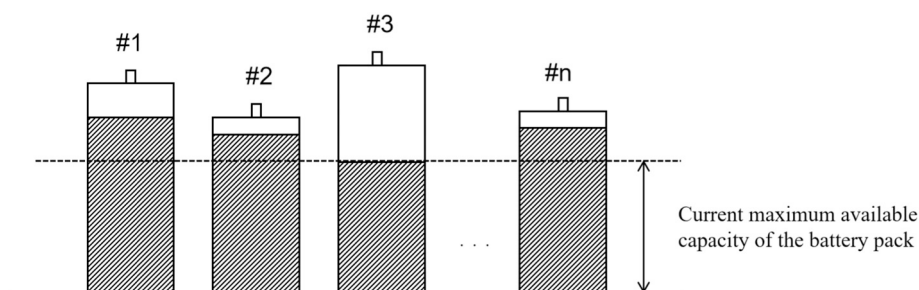


Fig. 3. Lithium-ion battery pack inconsistency.

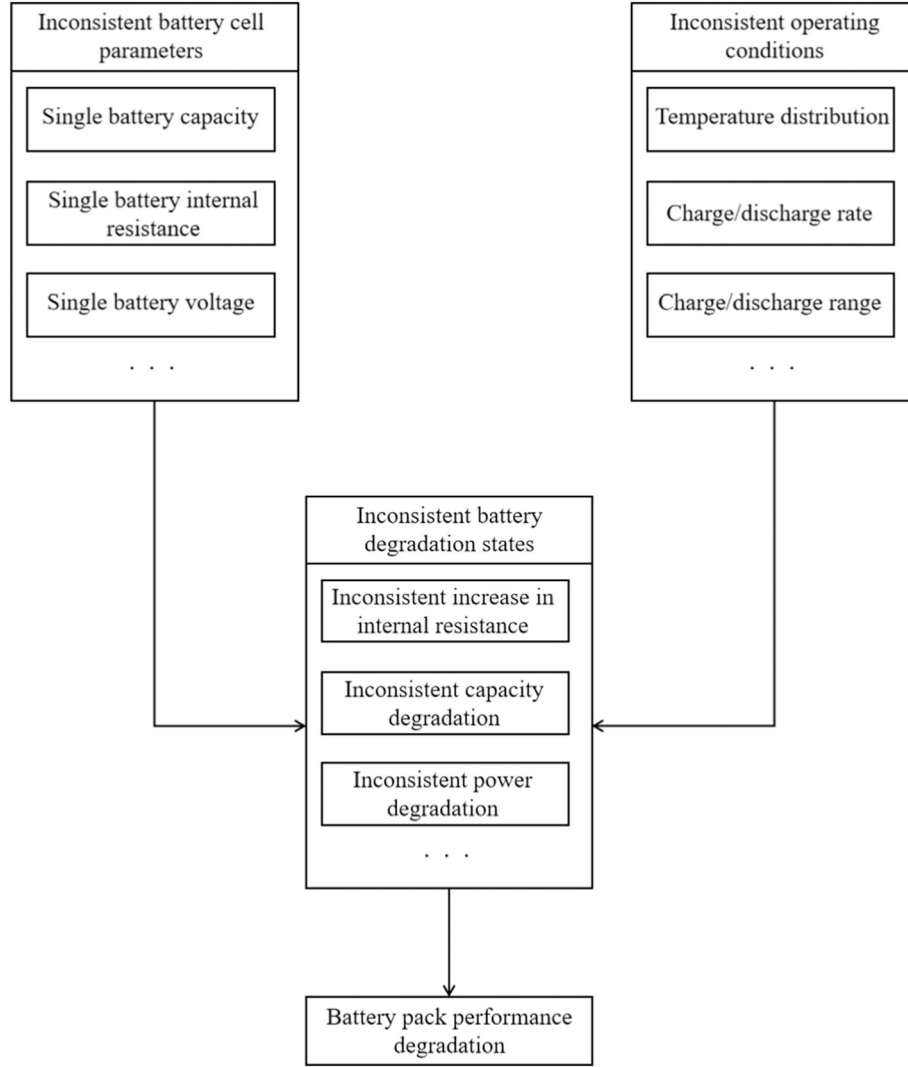


Fig. 4. Degradation mechanisms of lithium-ion battery packs.

effectively tracked and quantified [38].

Through examining the degradation mechanisms of lithium-ion battery packs, research indicates that their performance deterioration stems from two primary factors: individual cell degradation and increasing inconsistency between cells. These issues manifest in three key aspects: diminished capacity, heightened impedance, and decreased power output. Consequently, the evaluation of Li-ion battery pack degradation can be effectively characterized through available capacity, internal resistance, and power output, and based on the definition of the SOH of Li-ion battery packs, the assessment of performance degradation can be conducted directly. While direct assessment methods based on SOH definitions are feasible, practical limitations in data collection have prompted researchers to adopt alternative approaches. These include implementing incremental and statistical analyses using readily available parameters such as voltage, current, and temperature measurements. Additionally, researchers extract monotonic characteristic parameters over extended time periods to monitor degradation indirectly. This methodology proves advantageous as it eliminates the requirement for specialized testing conditions, thereby better reflecting actual operational scenarios and enabling more practical performance evaluation of lithium-ion battery systems.

Here, we will summarize the different health features used for model training and explore how to extract health features from five perspectives: incremental capacity analysis (ICA), time, frequency domain

analysis, multi-physical sensing technologies and battery pack inconsistency.

3.1. Incremental capacity analysis-based health feature

The Li-ion battery pack's performance deterioration will be directly evident in its charging and discharging procedures as a result of the individual cells' declining performance and the gradually increasing inconsistency among these cells. Thus, it is extremely effective to use the method of extracting the pertinent characteristics, like the battery pack's incremental capacity, in order to determine the degrading characteristics of lithium-ion batteries [39].

One aging characteristic parameter that is highly correlated and readily accessible online is the incremental capacity (IC) curve. The IC curves of batteries with different cycles are shown in Fig. 5. Each IC peak corresponds to a specific stage of the internal electrochemical reaction, with peak coordinates on the IC curve demonstrating strong correlation to battery capacity. As battery aging progresses, the IC curve exhibits predictable evolutionary patterns [40]. The IC can be mathematically expressed as follows:

$$\frac{dQ}{dV} \approx \frac{\Delta Q}{\Delta V} = \frac{Q_{t+\Delta t} - Q_t}{V_{t+\Delta t} - V_t} = \frac{\int Idt}{\Delta V}$$

where t , I , Q and V represent time, current, power and voltage,

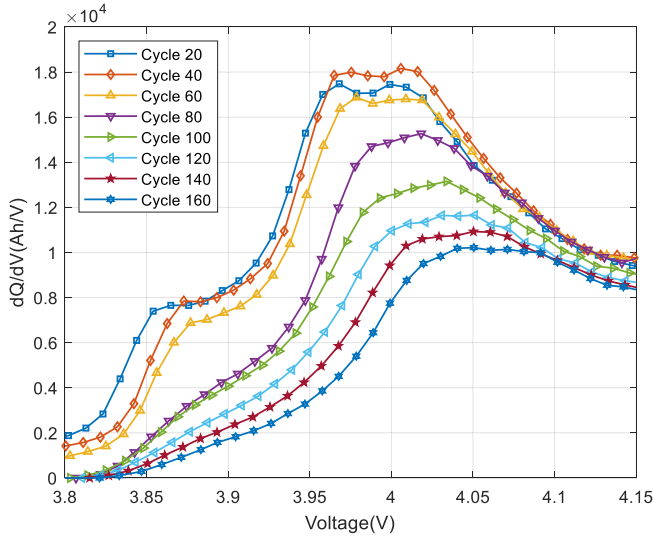


Fig. 5. Variation of IC curve of lithium-ion batteries at different cycles.

respectively.

Ye et al. [41] derived fusion health indicators (HIs) from IC curves using principal component analysis. They analyzed IC peaks to identify potential HIs, and developed a Gaussian process regression model with 95 % confidence intervals for high-precision SOH estimation based on principal component analysis utilizing fused HIs. Jiang et al. [42] developed a rapid IC curve acquisition technique, which they employed to create an adaptive capacity estimation framework that considers charging conditions. To enhance computational performance, a Kalman filter is used to quickly obtain smooth IC curves, requiring a reduced data length for processing the IC values. Additionally, to investigate how temperature, battery aging state, and starting charge state affect the IC curve, several conventional and non-standard battery charging tests were planned and conducted. Three characteristics of the IC curves during standard charging were identified to examine the correlation between feature height and battery capacity. Based on this analysis, an adaptive adjustment method for capacity estimation that takes into account the initial state of charge during charging was devised. Validation results confirm the efficacy of this correction method, emphasizing its precision and robustness. Huang et al. [43] used a data reconstruction method to obtain the change in IC curves with battery aging, and proposed a HI containing multi-perspective features for a more comprehensive characterization of battery degradation.

Nevertheless, it is crucial to acknowledge that the IC curve, derived from voltage and charge measurements, exhibits particular sensitivity to data acquisition noise, especially regarding peak characteristics such as amplitude and positional attributes [44,45]. There is a high demand for the acquisition accuracy and resolution of the state acquisition system. Therefore, how to realize the accurate taking and high-precision fitting of IC curves in various application scenarios, and accurately extract the deterioration traits of Li-ion battery performance of Li-ion battery packs contained in the curves is a major challenge for this method.

3.2. Time-based health feature

While IC analysis can provide an intrinsic mechanism explanation for the aging behavior, the data must be differentiated and filtered when obtaining the IC curves, which complicates data processing. As the battery ages, the charging time during the constant voltage phase gradually increases, while the charging time during the constant current phase gradually decreases in the constant current-constant voltage (CC-CV) charging process, which is commonly used for lithium-ion batteries. As a result, the variation in charging time during the CC-CV process of

Li-ion battery packs can also be utilized as a degradation characteristic of the packs.

Liu et al. [46] identified and extracted ten distinct health factors that comprehensively characterize the performance degradation patterns in lithium-ion batteries during CC and CV charging phases. Through careful analysis and implementation of these health factors, they successfully developed sophisticated models for tracking and predicting performance degradation in lithium-ion satellite power systems. In their analysis of the CV charging process, Wang et al. [47] conducted detailed investigations that revealed significant correlations between SOH and charging current profiles specifically during the CV phase. Building upon this research, they established new and effective SOH indicators, their methodology demonstrated that CV phase analysis alone could provide reliable health assessment metrics. Since it is difficult for an electric vehicle to appear to use the battery from a fully charged state to an empty state in a real use scenario, Zhang et al. [48] conducted a study to extract HIs from some constant voltage charging profiles. First, it was discovered through analysis of the CV charging curves that the CV charging curves may be stated in the same functional form in theory for all battery aging phases. The CV charging curve from the first cycle is therefore referred to as the original curve and the CV charging curves from subsequent cycles are regarded as fitted curves, depending on the findings of the CV charging curve analysis. The determination coefficient between these two curves is computed using the initial-stage CV charging data, which is referred to as the HI.

Time-based degradation state identification methodology demonstrates particular effectiveness in monitoring Li-ion battery charging processes, with its distinctive advantage lying in the straightforward computational approach to health feature extraction. Nevertheless, this analytical framework operates under specific constraints regarding charging protocols. Specifically, the methodology requires strict adherence to either CC or CV charging regimes throughout the entire process. The established mathematical models may lose their validity and predictive capabilities under scenarios involving operational variations, such as current magnitude adjustments during CC charging operations or voltage level modifications in CV charging phases. These inherent protocol restrictions and operational constraints significantly impact the method's versatility and widespread applicability across diverse charging scenarios, thereby introducing notable limitations to its practical implementation in real-world battery management systems.

3.3. Frequency domain analysis-based health feature

The degradation characteristic identification method based on frequency-domain features primarily evaluates battery health status through the analysis of battery characteristics in the frequency domain. This methodology predominantly employs Electrochemical Impedance Spectroscopy (EIS) technology, which has been extensively implemented in battery SOH prediction. EIS operates by applying small-amplitude alternating current signals and measuring the battery's voltage response, enabling non-invasive characterization of changes in internal electrochemical kinetic processes. It serves as a crucial tool in frequency-domain analysis for identifying battery aging characteristics. The spectral data encompasses battery response information across various frequencies, reflecting internal electrochemical processes, including parameters such as charge transfer impedance and double-layer capacitance [49]. These impedance spectra undergo significant changes as the battery experiences usage and aging. Fig. 6 illustrates the EIS of a battery.

Zhang et al. [50] developed an SOH prediction model by constructing a Simplified Equivalent Circuit Model (SECM) through mid-to-low frequency EIS, selecting parameters strongly correlated with battery SOH, and incorporating environmental temperature and dynamic charging SOC using a Gated Recurrent Unit (GRU) neural network. Zhang et al. [51] extracted SOH-related characteristic parameters, including double-layer impedance, SEI impedance, and contact

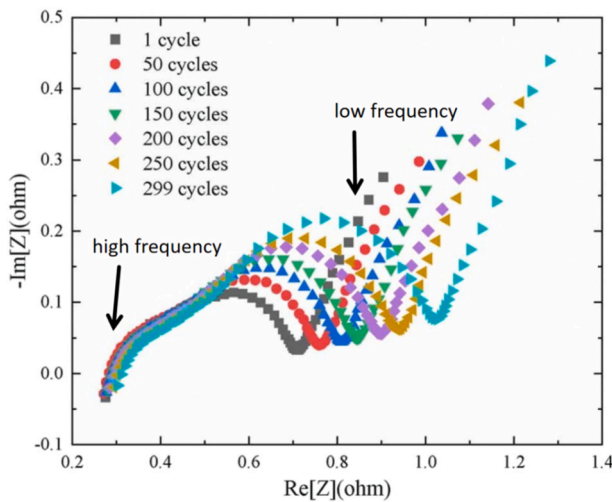


Fig. 6. Variation of EIS data of lithium-ion batteries at different charging and discharging cycles.

impedance, through distribution of relaxation times (DRT) analysis of EIS data. Li et al. [52] utilized the negative imaginary impedance in the EIS mid-frequency range as input features, capturing its significant increase during battery aging. Experimental results demonstrated that this method maintains high robustness under dynamic temperature conditions without relying on historical charge-discharge data. Pang et al. [53] constructed a quadratic polynomial model for SOH prediction by measuring the ratio between the low-frequency turning point and high-frequency zero-point real impedance in the EIS Nyquist plot. Results indicated that while this ratio remains insensitive to SOC variations, it decreases monotonically with capacity degradation. Moreover, compared to full-range EIS measurements, this method reduces measurement time by a factor of ten, making it suitable for rapid online detection, particularly in electric vehicle battery health assessment where real-time monitoring is crucial.

As a non-destructive testing methodology, EIS identifies characteristics of various physicochemical processes within batteries by measuring impedance responses across different frequencies. The degradation characteristic identification method based on EIS primarily characterizes battery performance deterioration through equivalent circuit fitting and characteristic parameter extraction. Its advantages include non-invasiveness, high sensitivity, early detection of microscopic degradation, and provision of multidimensional state information. However, its widespread application in real-time online monitoring is constrained by high equipment costs, lengthy measurement times, high environmental sensitivity, and demanding data processing and analysis requirements.

3.4. Multi-physical sensing technologies-based health feature

The complexity of battery degradation mechanisms often makes it challenging to comprehensively reflect the SOH using a single measurement approach. In recent years, the synergistic application of multi-physics measurement techniques has provided new insights into battery degradation mechanism research and SOH prediction [110]. Traditional SOH prediction methods typically rely on indirect characterization through single physical quantities (such as voltage or current). However, battery aging is fundamentally a complex process involving electrochemical-mechanical-thermal coupling, making it difficult for single parameters to comprehensively reflect internal state changes. Multi-physics sensing technologies, through the integration of temperature, pressure, deformation, and acoustic sensors, can accurately capture multidimensional information during battery aging, thereby enhancing prediction accuracy and robustness.

The combination of Electrochemical Impedance Spectroscopy and Acoustic Emission (AE) measurement represents a typical multi-physics measurement approach [111]. Davies et al. [112] achieved high-precision SOH prediction with average errors around 1 % by combining ultrasonic signal Time of Flight (ToF) shift, amplitude, and waveform with voltage measurements using an SVR model. This method requires only one ultrasonic measurement at full charge state to predict SOH, eliminating the need for historical data or cycling records, making it suitable for real-time monitoring in practical battery management systems. Chen et al. [113] established an electrochemical-thermal-aging effects coupling model by analyzing SEI film growth, crack propagation, and double-layer changes during battery aging. Through Pearson Correlation Coefficient (PCC) analysis, they discovered strong linear relationships between these parameters and SOH. Subsequently, these parameters were utilized as health features to develop SOH prediction models using four algorithms: Linear Regression (LR), Back Propagation Neural Network (BPNN), SVM, and GPR. Among these, BPNN and SVM demonstrated superior prediction performance with RMSE below 0.5 % and maximum errors under 1 %. Son et al. [114] extracted nine health indicators from mechanical and electrochemical measurements during the feature extraction phase, including multi-physics characteristics such as battery expansion/compression force, voltage, current, and impedance. In the feature processing phase, they employed robust weighted polynomial regression to eliminate outliers and used autoencoder neural networks for feature denoising, ultimately combining these with GPR to achieve SOH prediction.

3.5. Battery pack inconsistency-based health feature

Inconsistency of monomer within a Li-ion battery pack is one of the important mechanisms leading to degradation of battery pack performance [54,55]. Quantitative modeling of monomer inconsistency within a battery pack allows for a more accurate understanding of the overall performance characteristics of the battery pack and improves the ability to identify degraded states of the battery pack. Such quantitative modeling can be based on various parameters and characteristics of battery monomer, such as capacity, internal resistance, self-discharge rate, Coulombic efficiency, etc. [56], which can reveal the variability among different monomer within the battery pack.

Song et al. [57] conducted a numerical analysis of inconsistencies within these assemblies. They proposed a novel degradation feature for modeling Li-ion battery pack health. Their investigation examined the charging cutoff voltage inconsistency of each lithium-ion cell in the context of actual operation, with findings indicating that this inconsistency is prevalent among individual units and follows a normal distribution. Additionally, their research revealed that the scale parameter of the normal distribution demonstrates a negative correlation with the deterioration of overall pack performance. Fan et al. [58] investigated both series and parallel configurations through simulations. Their results demonstrated that current distribution inconsistencies decrease with more series connections, while increasing parallel connections and relative standard deviation of parameter distribution amplify these variations. Schuster et al. [59] measured the capacity and impedance of the 484 new and 1908 old lithium-ion batteries in two electric vehicles with the same batteries (954 batteries each) were characterized, and the results demonstrated that, as lithium-ion batteries aged, their various parameters transitioned from a normal distribution to a Weibull distribution. According to the research by Xia et al. [60], variations in internal resistance lead to uneven temperature distribution among individual cells within series-connected battery packs, potentially impacting the overall performance of these assemblies. In their comprehensive evaluation methodology for battery inconsistency, Duan et al. [61] developed an information entropy-based approach utilizing three distinct criteria: capacity measurements, internal resistance values, and the ratio between constant current charging capacity and constant voltage charging capacity.

Table 1
Comparison of different degradation feature recognition methods.

Method	Scope of application	Advantages	Limitations
Incremental capacity analysis-based health feature	Lithium-ion battery Lithium-ion battery packs	- Based on measurable voltage and current parameters, degradation feature extraction can be achieved; - It can be used simultaneously to analyze the degradation mechanism of lithium-ion battery monomers	- Highly sensitive to noise in voltage and current parameter measurements - Low-rate charging and discharging of battery or battery pack
Time-based health feature	Lithium-ion battery Lithium-ion battery packs	- Based on measurable voltage and current parameters, degradation feature extraction can be achieved - Effective degradation features can be extracted throughout the entire process of both the battery cell and the battery pack	- The extraction of degradation features during the discharge stage requires a constant discharge current, which is difficult to meet in practical applications - Some parameters require control of the battery's status
Frequency domain analysis-based health feature	Lithium-ion battery Lithium-ion battery packs	- It is capable of non-invasively characterizing changes in electrochemical kinetic processes within a battery - Information-rich and highly sensitive	- High equipment costs and time-consuming measurements - High requirements for data processing and analysis
Multi-physical sensing technologies-based health feature	Lithium-ion battery Lithium-ion battery packs	- Complementary information and strong robustness - Comprehensive data and real-time monitoring of batteries	- System complexity and difficulty in data processing - Higher cost and lower reliability due to use of multiple sensors
Battery pack inconsistency-based health feature	Lithium-ion battery packs	- It can directly characterize the performance degradation state of battery packs and respond to the performance degradation mechanism of battery packs - The analysis results are beneficial for continuing the individual equilibrium within the battery pack	- Evaluation of inconsistent available parameters is difficult to identify online - There is a lack of effective methods for the convergence of inconsistency evaluation results

One of the primary reasons for the decline in battery pack performance is increased battery monomer inconsistency. Although identifying degraded states of lithium-ion battery packs based on monomer inconsistency is an effective approach, several challenges exist. First, there is a wide variety of parameters for evaluating the battery monomer inconsistency, yet it is difficult to synthesize the inconsistency results between different parameters for evaluation. Therefore, how to effectively synthesize the inconsistency of multiple parameters is crucial for accurately evaluating battery pack performance degradation. Secondly, at this stage, the parameters used to evaluate the inconsistency of battery are mainly internal battery parameters, such as open-circuit voltage, ohmic internal resistance, direct current internal resistance, capacity and so on. However, the measurement and identification of these parameters also face certain challenges. Therefore, more reliable and accurate measurement methods need to be developed to meet the needs of battery monomer inconsistency evaluation in various application scenarios.

As shown in Table 1, it is a comparison and summary of the above five types of feature degradation recognition methods, which visually illustrates the scope of application and advantages and disadvantages of the five types of methods.

4. Lithium-ion battery pack SOH estimation

The definition of SOH for lithium-ion batteries and battery packs is essentially the same, it represents the ratio of the current performance of the battery pack to its performance at the beginning [62]. Two indicators are typically used to assess the SOH of lithium-ion batteries and battery packs: capacity and internal resistance:

- (1) From a capacity perspective [63]:

$$SOH = \frac{Q_{\text{present}}}{Q_{\text{new}}} \times 100\%$$

where Q_{present} and Q_{new} represent the maximum capacity of the battery pack under the current cycle and the maximum capacity of the new battery pack, respectively.

- (2) From an internal resistance perspective [64]:

$$SOH = \frac{R_{\text{EOL}} - R_{\text{present}}}{R_{\text{EOL}} - R_{\text{new}}} \times 100\%$$

where R_{EOL} , R_{new} and R_{present} represent the internal resistance of the battery pack under the current cycle at end-of-life, at new battery, and at the current cycle, respectively.

However, in applications such as electric vehicles, the state of energy assumes a more significant role. Compared to the internal resistance measured by methods such as EIS [65] and hybrid pulse power characterization (HPPC) [66], capacity can be directly accumulated through ampere-hour integration, making it a more practical indicator for assessing SOH.

The methods for estimating SOH in lithium-ion battery packs can be categorized into three fundamental approaches based on their modeling principles and theoretical foundations: filter-based models, machine learning-based models, and hybrid estimation models.

4.1. Statistical filter-based SOH estimation models

Statistical filter based methods play an important role in the SOH estimation modeling of lithium-ion batteries. This method is usually used to realize the optimal estimation of battery parameters or SOH by establishing a state-space model, representing the state of the battery, such as state of charge (SOC), SOH, as a state variable, and utilizing systematic observations such as voltage, current, temperature, etc. [67]. Popular statistical filter algorithms include the particle filter and its extension techniques, as well as the Kalman filter and its methods of extension.

4.1.1. Kalman filter

The Kalman filter, a recursive Bayesian filtering algorithm, enables state estimation in linear dynamic systems [68,69]. This computational approach offers advantages of low processing complexity and rapid model updating, making it suitable for estimating both SOC and SOH in battery management systems. However, given its primary application to linear systems, modifications such as Extended Kalman Filter (EKF) or Unscented Kalman Filter (UKF) are necessary when addressing strongly nonlinear systems like batteries [70].

Erlangga et al. [71] implemented dual Kalman filter (DKF) methodology for real-time estimation of lithium battery SOC and SOH. Yang et al. [72] created a parameter-inconsistent electrical simulation model for Li-ion batteries that took thermal effects into account. The model was based on the double extended Kalman filter (DEKF) algorithm and the

second-order resistor-capacitor equivalent circuit model, which calculates temperature through the thermal module and estimates SOC and SOH using the DEKF. Shu et al. [73] estimated the battery's state by incorporating temperature corrections using an adaptive EKF with a forgetting factor. Xu et al. [74] utilized a multi-scale DEKF to estimate the battery's SOC and SOH.

4.1.2. Particle filter

The particle filter utilizes Monte Carlo methods to approximate the a posteriori probability distribution of the system's state through a collection of random particles, allowing it to accommodate various types of state-space models and effectively tackle nonlinear and non-Gaussian issues [75].

Schwunk et al. [76] proposed a particle filter method using Monte Carlo sampling to address the flat and ambiguous open-circuit voltage characteristics of lithium iron phosphate/graphite batteries. The method can represent arbitrary probability density functions. The ambiguity can be modeled stochastically, which avoids dealing with complex models with hysteresis. Bi et al. [77] proposed a method for SOH estimation of battery packs based on the genetic resampling particle filter method after simplifying the process of modeling and the problem that the measurement process contains multi-source noise, which leads to the system presenting a non-Gaussian nature. Shi et al. [78] proposed an improved unscented particle filter(IUPF) that generating important probability density functions using UKF and proposed a method to select the most suitable particles at the resampling stage.

4.2. Machine learning-based SOH estimation models

In machine learning-based SOH estimation, Li-ion batteries function as a 'black box' [79]. This approach eliminates the need for detailed knowledge of specific operating environments or internal degradation mechanisms, instead relying on algorithmic data processing [80]. The methodology leverages historical monitoring and offline testing data to establish correlations between deterioration characteristics and key metrics such as internal resistance and capacity [81]. Through extensive analysis of historical data, the system automatically identifies parameters and learns association patterns. Current battery conditions are evaluated by inputting degradation state identification results, which enable SOH estimation through analysis of capacity and internal resistance characteristics. This data-driven framework's key advantage lies in its ability to automatically learn and refine its model through continuous exposure to experimental data, progressively enhancing estimation accuracy and reliability.

4.2.1. Support vector machine (SVM)

SVM, a widely adopted machine learning approach illustrated in Fig. 6, has proven particularly effective in developing data-driven SOH models for both individual lithium-ion batteries and battery packs. The core principle involves identifying a hyperplane within high-dimensional space to effectively separate distinct data categories for classification or regression tasks [82]. In the context of Li-ion battery health estimation, the model learns an optimal hyperplane through analysis of historical monitoring data and offline test results. This trained hyperplane can then effectively evaluate and predict battery SOH based on parameters including internal resistance and capacity.

Patil et al. [83] analyzed Li-ion battery cycling data under varied operating circumstances, which allowed for the extraction of important characteristics from the voltage and temperature profiles. They created SVM-based classification and regression models using these essential features. Yang et al. [84] enhanced support vector regression (SVR) by integrating particle swarm optimization (PSO) to optimize variable parameters. Their resulting PSO-SVR model for state of health (SOH) estimation demonstrated robust accuracy, achieving root-mean-square errors of 1.59 % and 0.56 % across two battery groups. Feng et al. [85] introduced an online machine learning approach for lithium-ion

battery SOH estimation. Their methodology first identifies support vectors representing intrinsic battery characteristics from new unit charging data. These vectors subsequently enable determination of SVM coefficients for batteries at varying health states. The system functions by comparing partial charge curves with stored SVM data, ultimately quantifying SOH through similarity factor analysis.

4.2.2. Relevance vector machine (RVM)

RVM is a sparse probabilistic model based on Bayesian theory [86]. RVM can naturally generate sparse solutions through Bayesian inference, which makes it use typically very few correlation vectors, thus simplifying the model and reducing computational cost. Additionally, while SVM cannot assess the uncertainty of the projected outcomes, RVM is capable of doing so.

Song et al. [87] developed a nonlinear mapping system between degradation features and battery capacity using RVM, which provided quantitative estimates alongside their error margins. Dong et al. [88] introduced a hybrid kernel RVM (H-RVM) methodology for SOH assessment. Their dual-RVM approach incorporates one model for lithium battery aging patterns, while another performs multi-step predictions of the Indirect Health Factor (IHF). Through optimization of kernel parameters, Chen et al. [89] developed BA-RVM, a system that enhances the relevance vector machine using the bat algorithm, resulting in improved SOH estimation precision.

4.2.3. Gaussian process regression (GPR)

GPR functions as a nonparametric model for data regression analysis, leveraging a Gaussian process prior [90]. This approach distinguishes itself through its generalizability and resolvability, offering both point estimates and posterior probability distributions of predicted outcomes. Such capabilities make it particularly valuable in lithium battery SOH estimation, as it provides both predicted values and corresponding confidence intervals, enabling comprehensive assessment of prediction reliability and accuracy.

Deng et al. [91] enhanced the conventional GPR framework by developing an autoregressive model to improve both estimation accuracy and confidence levels. Zhang et al. [92] created a precise prediction methodology by combining EIS with Gaussian process machine learning. Their approach utilizes the complete spectrum as input, with the model automatically identifying spectral features indicative of degradation, thus eliminating the need for feature engineering. Integrating classical convolutional neural networks (CNN) with GPR modeling, Buchanan et al. [93] developed a comprehensive system that combines feature identification and many-to-one mapping capabilities with probabilistic analysis. Their approach processes random partial charge segments to predict battery SOH with an average absolute error below 1 %, while leveraging GPR variance to quantify prediction confidence.

4.2.4. Neural network (NN)

The NN is connected by a multitude of neurons and adjustable connection weights between them, which makes it capable of handling massively parallel processing, self-organizing and self-learning, and flexibility [94]. It is due to these features that neural networks excel in handling complex nonlinear problems and hence can be applied to determine a battery's SOH. In Fig. 7, the NN prediction framework is displayed.

Kheirkhah-Rad et al. [95] leveraged deep feedforward neural networks to evaluate 90 distinct shallow and deep architectures for characterization. These frameworks underwent deployment, training, and testing across 17 batteries utilizing various cycling techniques. Their findings revealed that individual test units exhibited errors below 1 %, achieving an overall root mean square percentage error of 0.43 %. Zhang et al. [96] implemented a feature extraction methodology combining principal component analysis and variable correlation analysis, generating a compressed dataset that maintained strong correlations with charging states of battery packs. This processed information

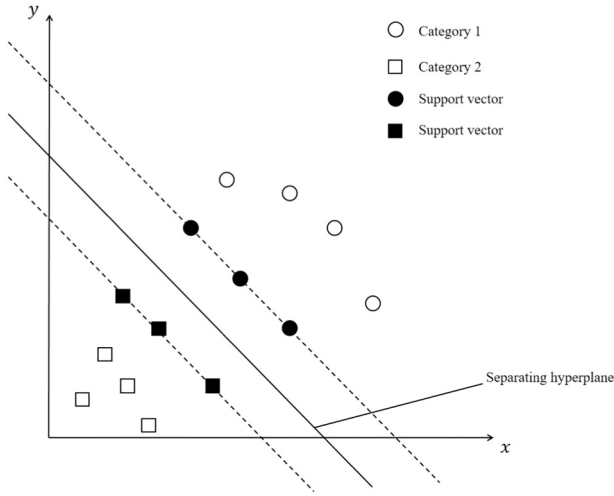


Fig. 7. Principle diagram of SVM.

was subsequently channeled into a radial basis function neural network (RBFNN) for state estimation. Shen et al. [87] engineered a state of health (SOH) assessment model utilizing deep convolutional neural networks (DCNN). Recognizing the constraints of limited training samples in real-world applications, they integrated the principles of transfer learning and ensemble learning. Their approach focused on enhancing model accuracy through the incorporation and transfer of diverse parameter sets. The research by Bockrath et al. [98] introduced an innovative algorithm for SOH determination based on partial discharge

curves from different segments. Their methodology directly feeds raw sensor data into a temporal CNN, which processes the information automatically. This approach enables SOH assessment under various aging and degradation scenarios without requiring a feature engineering phase.

4.2.5. Extreme learning machine (ELM)

ELM, proposed by Huang et al. [99], represents an efficient training algorithm for single-hidden-layer neural networks. Its distinctive feature lies in the random assignment of weights and biases between the input and hidden layers, eliminating the need for adjustments during training or iterative steps [100]. This approach significantly streamlines parameter determination while achieving superior speed and accuracy compared to conventional learning algorithms. The architectural framework of ELM is illustrated in Fig. 8.

Building upon the basic ELM framework, researchers have developed various enhanced approaches. Fu et al. [101] used the ELM with regularization mechanism to establish a SOH evaluation model after screening the health factors. Zhao et al. [102] developed a hybrid PSO-ELM model by employing particle swarm optimization to fine-tune the weights and biases following health feature parameter selection. Zhang et al. [103] extended this PSO-ELM framework to simultaneously estimate both SOH and RUL. Meanwhile, Pan et al. [104] focused on improving online estimation efficiency by utilizing ELM to establish correlations between extracted HIs and capacity deterioration (Fig. 9).

4.3. SOH estimation based on hybrid methods

Hybrid health state estimation refers to the assessment of a battery's SOH through the integration of two or more methodologies. Such

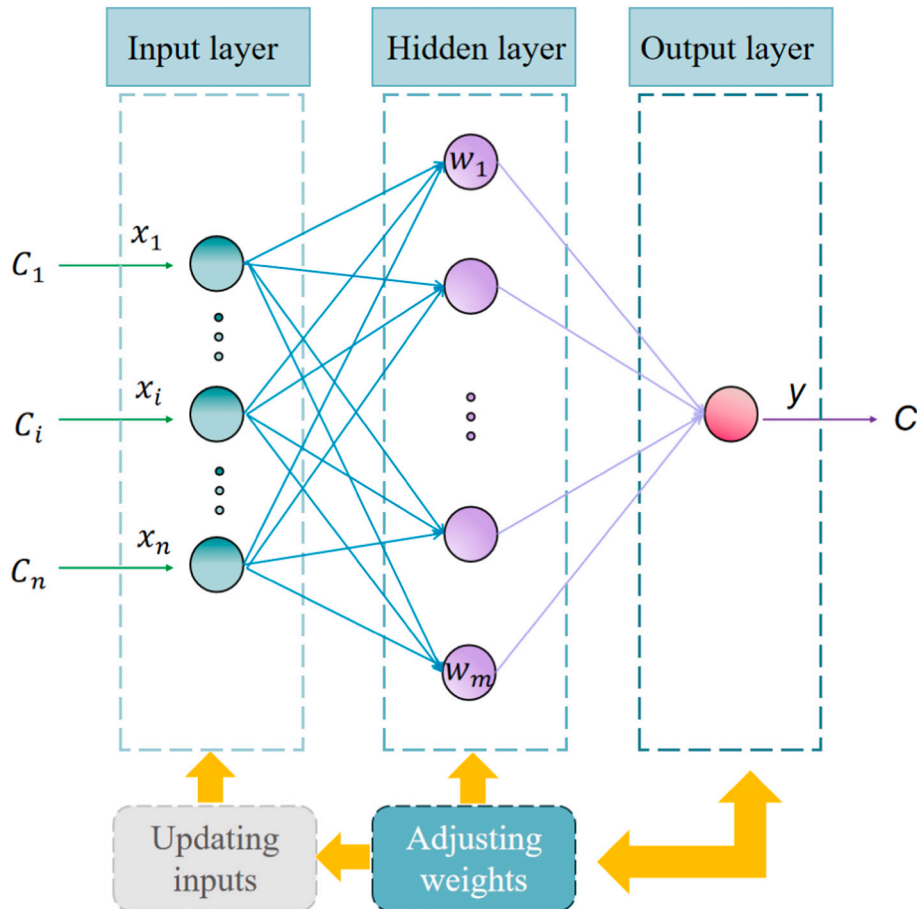


Fig. 8. The structure of NN.

combinatorial approaches enable the effective utilization of distinct advantages from different methods. Park et al. [105] developed an innovative framework that integrates DEKF with autoregressive models to evaluate SOH through statistical modeling, particularly beneficial when measurement data is scarce. While DEKF excels in real-time battery state estimation, its capacity estimation capabilities are limited; thus, a multivariate autoregressive model was incorporated to enhance the prediction accuracy. He et al. [106] engineered a sophisticated system combining LSTM and GRU networks, capitalizing on LSTM's exceptional capability in processing high-dimensional data while leveraging GRU's proficiency in handling extensive data volumes. Meanwhile, Li et al. [107] introduced a comprehensive model designed for dual purposes: evaluating short-term health status and forecasting long-term residual service life of batteries. Their methodology conducts thorough analysis of partial incremental capacity to identify four crucial variables, subsequently integrating nonlinear regression with GPR for SOH prediction. Deng et al. [108] proposed an advanced battery capacity prediction framework based on the synergy between Seq2Seq and GPR models. This approach processes electric vehicle charging data to obtain battery pack capacity measurements and relevant feature sets for model training. The methodology underwent rigorous validation using data collected from 20 electric vehicles over approximately 29 months, achieving impressive accuracy metrics with mean absolute error and root mean square error of 1.24 % and 1.53 % respectively. In a separate investigation, Li et al. [109] developed a sophisticated hybrid methodology combining Gaussian process modeling with particle filtering techniques to estimate SOH in lithium-ion batteries under uncertain conditions. This innovative approach demonstrates the ability to learn degradation process distributions utilizing solely the available capacity monitoring data.

The three types of SOH estimate methods for lithium-ion battery packs are compared and summarized in Table 2, which provides a visual representation of the approaches' respective applications as well as their

benefits and drawbacks. Meanwhile, Table 3 summarizes the comparison of parameters and their effectiveness among various estimation methods.

5. Challenges and future prospects

5.1. Problems and challenges

Although significant progress has been made in estimating the SOH of lithium-ion battery packs in recent years, several challenges and issues remain to be addressed.

(1) How to accurately measure SOH-related parameters?

By measuring certain parameters inside the battery (e.g., internal resistance, impedance, etc.) it is possible to reflect the state health of the Li-ion battery to a certain extent and reveal the electrochemical processes of its interactions. Nevertheless, because of the reciprocal coupling between the battery's internal reactions, there is often uncertainty in the measurement data. Furthermore, in addition to the intrinsic elements, one of the main concerns that greatly affects the accuracy of the final results is the sensor's accuracy. Therefore, during the actual assessment of battery SOH, the parameters measured by the assessment method may be influenced by these factors, potentially resulting in significant errors.

(2) How to improve SOH estimation accuracy with poor training data?

Contemporary battery pack SOH assessment models incorporate multiple electrical parameters as key indicators, including variations in current flow, voltage dynamics, impedance characteristics, and incremental capacity profiles. These measurement-based metrics form the foundation for evaluating battery degradation patterns and predicting remaining useful life.

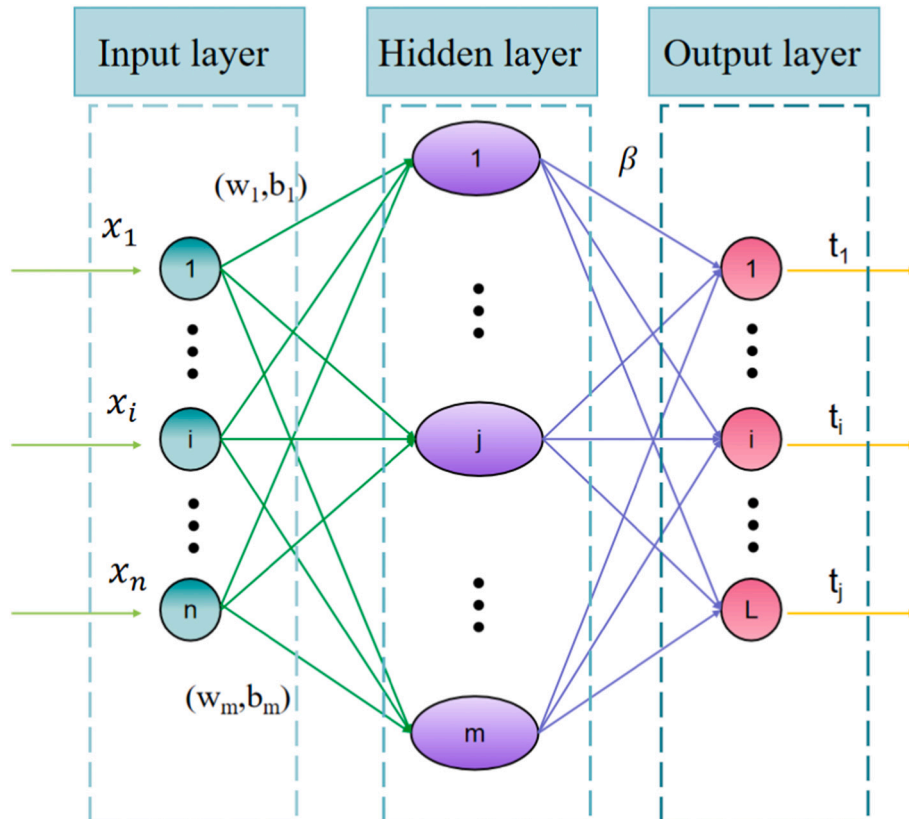


Fig. 9. The structure of ELM.

Table 2

Comparison of advantages and disadvantages of different SOH estimation methods.

Method	References	Advantages	Disadvantages
Statistical filter-based SOH estimation models	Kalman filter [71–74]	Low computational complexity and fast model update speed	High requirement for measurement accuracy and not suitable for nonlinear systems.
	Particle filter [76–78]	Wide applicability and high flexibility	Complex computation and reliance on resampling
Machine learning-based SOH estimation models	SVM [83–85]	High accuracy and fast calculation speed	Difficulty in hyperparameter selection, issues with kernel function choice, and lack of output confidence intervals
	RVM [87–89]	Easy parameter tuning and strong noise resistance	Slow computation speed and sensitivity to outliers
	GPR [91–93]	High accuracy and output with probability distribution	High computational complexity and poor online update capability
	NN [95–98]	High accuracy and easy modeling	High computational complexity, limited accuracy with small sample data, and inability to provide real-time estimation
	ELM [101–104]	Fast computation speed, short processing time, and certain generalization capability	Low accuracy, no confidence interval output, and poor robustness.
SOH estimation based on hybrid methods	[105–109]	High accuracy and strong generalization capability	Complex computations

However, in practice, it is not possible to obtain a data set that is as accurate as the one in the laboratory due to the influence of the use environment or the limitation of the sensor accuracy at the time of use. To address this practical constraints, the selection of both appropriate measurement parameters and their subsequent processing methodologies becomes critical. This involves implementing robust filtering techniques, adopting adaptive sampling methods, and utilizing advanced algorithmic approaches to extract meaningful insights from potentially noise-contaminated signals.

(3) How can the applicability of SOH estimation methods be improved?

Current SOH estimation approaches primarily operate under controlled laboratory conditions, such as stable temperature, voltage, and current charging environments. These conventional methods have demonstrated satisfactory performance in experimental settings. However, real-world applications present significantly more complex scenarios, where multiple environmental factors fluctuate simultaneously. Particularly challenging is the variability of ambient temperatures, which can dramatically affect battery behavior and render traditional estimation methods less reliable. Additionally, factors such as charging patterns, discharge rates, and usage cycles often deviate from ideal laboratory conditions. This disparity between controlled testing and actual operating conditions highlights the urgent need to enhance the adaptability of SOH estimation techniques. Future research should focus on developing robust algorithms that can maintain accuracy across diverse environmental conditions while accommodating the dynamic nature of practical applications.

5.2. Future prospects

Although our methods for predicting the SOH of Li-ion battery packs have developed relatively well, there are still the following shortcomings:

(1) Development of an innovative method for estimating SOH in lithium-ion batteries

Today's lithium-ion battery market is dominated by lithium cobaltate, lithium ternary and lithium iron phosphate batteries. All of them use organic solvents as electrolyte, which leads to combustion and explosion easily. Solid-state lithium-ion batteries use a solid electrolyte to replace the traditional liquid electrolyte, which fundamentally eliminates the potential for electrolyte corrosion and leakage, greatly reduces the risk of thermal runaway and explosion of the battery, and is considered to be the

ideal chemical power source for new energy storage and electric vehicles and other application scenarios. Therefore, modeling, experiments, and measurement methods for new lithium-ion batteries need to be improved.

(2) Optimization of SOH feature parameter extraction

The future evolution of SOH estimation techniques points toward two parallel development paths. The first focuses on identifying novel health indicators that correlate strongly with battery degradation while maintaining measurement feasibility. The second path explores advanced non-invasive diagnostic technologies. Ultrasonic measurement techniques can potentially reveal internal structural changes, while laser-based methods offer high-precision surface analysis. Infrared thermography enables detailed thermal mapping, providing insights into local degradation patterns and charge distribution anomalies. Compared to traditional measurement methods that face additional complexity due to the interaction between batteries, these emerging technologies have great potential in battery pack applications.

(3) SOH estimation method optimization

There are differences in complexity, computational efficiency, feasibility, and applicability among different SOH estimation methods, and it is a major trend in the future development of SOH to complement the strengths and weaknesses of different SOH estimation methods so that they can match the most suitable application conditions, for example, improving the existing data-driven methods or combining multiple methods so as to increase the estimation efficiency of SOH.

Furthermore, the optimization process should consider application-specific requirements. Different scenarios - from grid-scale energy storage to electric vehicles present unique challenges and operational constraints. Therefore, developing adaptable frameworks that can be fine-tuned for specific use cases while maintaining robust performance becomes crucial. This might involve adjusting model complexity based on available computational resources, optimizing algorithm efficiency for real-time applications, or incorporating domain-specific constraints into the estimation process.

(4) Battery management system (BMS) application development for SOH

The BMS functions as the central intelligence of the entire battery system, ensuring the safety, reliability, and longevity of the battery pack through comprehensive state monitoring, efficient management, and protective measures. Its role is particularly vital in electric vehicles, energy storage systems, and consumer electronics, where it simultaneously enhances battery performance and durability while mitigating potential safety

Table 3

The detailed synopsis of SOH prediction methods for battery packs.

Method	References	Health feature	Feature sources	Estimation error
Kalman filter	[71]	Internal resistance	Charging and discharging	RMSE = 1.2 %
	[72]	Maximum capacity and internal resistance of the battery cell	Charging and discharging	Error = 3.37 %
	[73]	Feature point voltage	Discharging	Error < 4.14 %
	[74]	Temperature, open circuit voltage, current	Charging and discharging	7.57 % < RMSE < 10.72 % (high current) 5.52 % < RMSE < 8.52 % (low current)
Particle filter	[76]	SOC	Charging and discharging	N/A
	[77]	Internal resistance	Charging	0.35 % < Variance < 0.46 %
	[78]	Internal resistance	Charging and discharging	0.75 % < RMSE < 1.16 %
SVM	[83]	Voltage and temperature profiles	Charging and discharging	RMSE = 0.357 %
	[84]	The peak values of IC curves	Charging	RMSE = 1.59 %
RVM	[85]	Partial charging curve	CC charging	Less than 3 % error at 95 %
	[87]	Voltage and current data of battery pack	Partial DOD	MAE = 0.11 % RMSE = 0.17 % STD = 0.19 %
	[88]	Integral of charging voltage curve, integral of charging current curve, and integral of charging temperature curve within a specific voltage range	Charging	MAE = 0.31 % MAPE = 0.32 % RMSE = 0.42 %
	[89]	Equal voltage drop discharge Time	Discharging	RMSE = 1.18 %
GPR	[91]	Battery current, pack voltage, cell voltage, and cell temperature	Charging and discharging	RMSE = 2.19 % MaxAE = 5.64 % R ² value is generally > 0.85
	[92]	EIS	Charging and discharging	
	[93]	Random partial charge segments	Random charging data	MAE = 0.6 %
NN	[95]	Referenced-based Charging Time	Charging and discharging	RMSPE < 1 %
	[96]	Maximum voltage and minimum voltage of battery cell, Average voltage of battery pack, battery pack voltage, circuit current, average temperature of battery	Charging and discharging	Error = 2.02 %
	[97]	Voltage curves, current curves, charge capacity curves	Charging	RMSE = 1.503 %
	[98]	Discharge current, discharge time	Discharging	RMSE = 1.0 % MAEP = 0.4 %
ELM	[101]	The peak value of the imaginary part, the real part at the peak of the imaginary part, the amplitude at the peak of the imaginary part, the imaginary part and the amplitude at 0.1 Hz, the phase peak value	Charging and discharging	0.55 % < Error < 1.12 %
	[102]	Voltage difference between cells, charging Time difference, accumulated mileage, driver's charging behavior with temperature deviations, SOC at the end of discharge	Charging and discharging	RMSE = 0.0849 % MAPE = 0.071 %
	[103]	The time it takes for the discharge voltage to drop from 3.8 V to 3.5 V, the average value of the battery voltage during discharge, the average value of the battery temperature during discharge, cycle number	Discharging	MAE = 0.26 % RMSE = 0.33 %
	[104]	Increase in the ohmic internal resistance and the polarized internal resistance	Charging and discharging	Error < 2.5 %
Hybrid methods	[105]	Ohmic resistance, diffusion resistance	Charging and discharging	MAPE = 0.4165
	[106]	Accumulated mileage, statistical characteristic distribution of discharge multiplicity, ambient temperature and SOC usage frequency number	Charging and discharging	MAE = 0.014 RMSE = 0.032
	[107]	Peak point, intercept, and slope of IC curve	Charging and discharging	Error < 2 %
	[108]	The average and standard deviation of current, the sum and standard deviation of battery pack voltage, the standard deviation of SOC, the average of battery voltage difference, the maximum temperature of the battery, and the sum of temperature difference	Charging and discharging	MAE = 1.24 % RMSE = 1.53 %
	[109]	Capacity	Charging and discharging	0.7938 % < RMSE < 1.7768 % 0.0087 % < MAPE < 0.0217 %

risks.

The core functionalities of BMS encompass state monitoring, safety protection, performance optimization, and predictive fault diagnosis. Among these, state monitoring focuses on three critical parameters: SOC, State of Power (SOP), and SOH. Battery aging leads to SOH degradation and capacity reduction, meaning that even at 100 % SOC, an aged battery delivers less energy than a new one. Moreover, the declining SOH increases internal resistance and reduces instantaneous power output capability,

consequently diminishing SOP. Thus, precise SOH estimation is fundamental to effective battery management.

Despite advances in Li-ion battery technology, significant challenges remain in bridging the gap between SOH prediction technologies and their practical implementation. While various SOH estimation methods exist, they often fall short in providing real-time, accurate, and widely applicable solutions that meet BMS monitoring requirements. Future integration of advanced SOH estimation with BMS holds promising potential: combining real-time monitoring technologies with

sophisticated data analysis algorithms could enable dynamic SOH estimation and prediction. This integration would facilitate adaptive management strategies, including optimized charging protocols and intelligent load control, ultimately enhancing system reliability, safety, and battery longevity.

6. Conclusion

The estimation of SOH for lithium-ion battery packs has become a significant concern across various industries, including energy storage, electric vehicles, and aerospace, due to the rapid adoption of lithium-ion batteries in these sectors, and a research hotspot for the intersection of electrochemistry, test and measurement, and data-driven and other multidisciplinary research. In this paper, we conduct a review study on the SOH estimation methods for Li-ion battery packs. This paper firstly introduces the working principle and degradation mechanism of batteries and battery packs, and introduces the degradation feature identification of lithium-ion battery packs from three aspects, namely, incremental capacity analysis based, time based and inconsistency based. Meanwhile, the current methods for Li-ion battery pack SOH estimation are elaborated and analyzed in detail from three aspects, namely, statistical filter based, Machine learning based and hybrid methods, and their advantages and disadvantages are fully compared. Finally, challenges and outlooks are presented from the current SOH estimation techniques. Some worthwhile research directions are discussed in terms of improving the measurement accuracy, improving the applicable range of SOH estimation methods, and optimizing SOH estimation methods. In summary, SOH estimation of battery packs will have a profound impact on several important areas in the future, and it will become one of the hot spots for future research. Moreover, the analytical comparisons and conclusions in this paper are of great significance for the future development of SOH estimation of lithium-ion battery packs, and we hope that they can provide reference and reference for researchers in related fields.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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